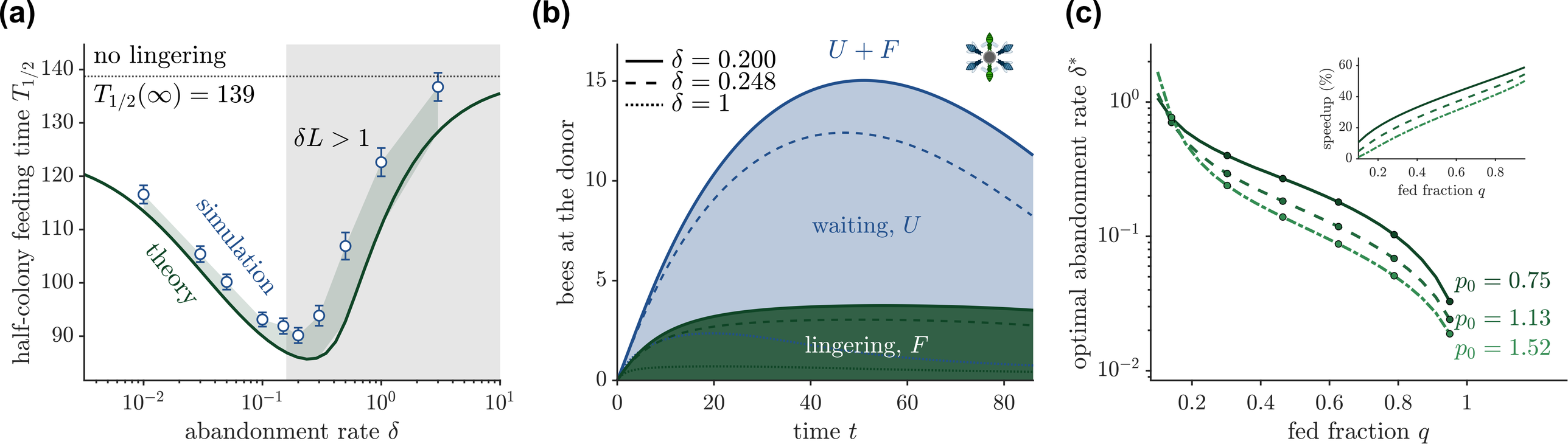

(a)
no lingering
$T_{1/2}(\infty) = 139$
$\delta L > 1$
simulation
theory
half-colony feeding time $T_{1/2}$
abandonment rate δ
(b)
$\delta = 0.200$
$\delta = 0.248$
$\delta = 1$
$U + F$
waiting, U
lingering, F
bees at the donor
time t
(c)
optimal abandonment rate δ^*
fed fraction q
speedup (%)
$p_0 = 0.75$
$p_0 = 1.13$
$p_0 = 1.52$